

## UNIVERSITY NAME

COURSE NAME — Final Exam

Year: 2025

## Exercise 1:

Let  $E = C^1([0, 1], \mathbb{R})$ , the space of  $C^1$  functions from  $[0, 1]$  to  $\mathbb{R}$ . For  $f \in E$ , define

$$N(f) = \|f\|_\infty + \|f'\|_\infty \quad \text{with} \quad \|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|.$$

1. Show that  $N$  is a norm on  $E$ . Is it equivalent to  $\|\cdot\|_\infty$ ?
2. (a) Verify that  $N'(f) = |f(0)| + \|f'\|_\infty$  is a norm on  $E$ .  
(b) Show that it is equivalent to  $N$ .

## Answer Area

## 1. Norm properties:

- **Positivity:**  $N(f) \geq 0$ . If  $N(f) = 0$ , then  $\|f\|_\infty = 0$  and  $\|f'\|_\infty = 0$ , so  $f = 0$ .
- **Homogeneity:**  $N(\lambda f) = |\lambda|(\|f\|_\infty + \|f'\|_\infty) = |\lambda|N(f)$ .
- **Triangle inequality:**  $N(f + g) \leq \|f\|_\infty + \|g\|_\infty + \|f'\|_\infty + \|g'\|_\infty = N(f) + N(g)$ .

Thus  $N$  is a norm.

**Equivalence with  $\|\cdot\|_\infty$ :** They are not equivalent. Consider  $f_n(x) = \sin(n\pi x)$ . Then:

$$\|f_n\|_\infty = 1, \quad \|f'_n\|_\infty = n\pi, \quad \text{so } N(f_n) \sim n\pi \rightarrow \infty$$

while  $\|f_n\|_\infty$  remains bounded. No constant  $C$  exists such that  $N(f) \leq C\|f\|_\infty$  for all  $f$ .

2. (a) Norm properties for  $N'$ :

- **Positivity:** Clear. If  $N'(f) = 0$ , then  $|f(0)| = 0$  and  $\|f'\|_\infty = 0$  so  $f' = 0 \implies f$  constant, and with  $f(0) = 0 \implies f = 0$ .
- **Homogeneity:**  $N'(\lambda f) = |\lambda f(0)| + \|\lambda f'\|_\infty = |\lambda|N'(f)$ .
- **Triangle inequality:**  $N'(f + g) \leq |f(0) + g(0)| + \|f' + g'\|_\infty \leq N'(f) + N'(g)$ .

- (b) **Equivalence with  $N$ :** Show  $\exists c, C > 0$  such that  $cN(f) \leq N'(f) \leq CN(f)$ .

- **Upper bound:**  $N'(f) = |f(0)| + \|f'\|_\infty \leq \|f\|_\infty + \|f'\|_\infty = N(f)$ .
- **Lower bound:** For any  $x \in [0, 1]$ ,  $f(x) = f(0) + \int_0^x f'(t)dt$ , so

$$|f(x)| \leq |f(0)| + \int_0^1 |f'(t)|dt \leq |f(0)| + \|f'\|_\infty.$$

Taking supremum:  $\|f\|_\infty \leq |f(0)| + \|f'\|_\infty = N'(f)$ . Thus  $N'(f) \leq N(f) \leq 2N'(f)$  (since  $N(f) = \|f\|_\infty + \|f'\|_\infty \leq N'(f) + \|f'\|_\infty \leq 2N'(f)$ ).

So  $N$  and  $N'$  are equivalent with constants  $c = 1/2$ ,  $C = 1$ .

## Exercise 2:

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be a continuous function and  $I$  an open interval in  $\mathbb{R}$ . Show that  $E = \{x \in \mathbb{R}^n : f(x) \in I\}$  is an open subset of  $\mathbb{R}^n$ .

### Answer Area

Let  $x_0 \in E$ . Then  $f(x_0) \in I$ . Since  $I$  is open, there exists  $\varepsilon > 0$  such that

$$(f(x_0) - \varepsilon, f(x_0) + \varepsilon) \subset I.$$

Since  $f$  is continuous at  $x_0$ , there exists  $\delta > 0$  such that for all  $x \in \mathbb{R}^n$  with  $\|x - x_0\| < \delta$ , we have

$$|f(x) - f(x_0)| < \varepsilon.$$

This implies  $f(x) \in (f(x_0) - \varepsilon, f(x_0) + \varepsilon) \subset I$ . Therefore, for all  $x$  in the open ball  $B(x_0, \delta)$ , we have  $f(x) \in I$ , i.e.,  $x \in E$ .

Thus, every point  $x_0 \in E$  has an open neighborhood contained in  $E$ , so  $E$  is open in  $\mathbb{R}^n$ .

## Exercise 3:

1. Calculate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x^2 + y^2}$$

2. Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be defined by

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{\sqrt{(x^2 + y^2)^3}} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

- (a) Show that  $\nabla f(0, 0) = (0, 0)$ .
- (b) Show that the function  $\frac{\partial f}{\partial x} : \mathbb{R}^2 \rightarrow \mathbb{R}$  is not continuous at  $(0, 0)$ .

3. Let  $U$  be an open subset of  $\mathbb{R}^2$ ,  $(u, v) \in U$  with  $u \neq 0$ , and  $f : U \rightarrow \mathbb{R}$  a  $C^1$  function, homogeneous of degree  $\alpha$ , such that:

$$f(u, v) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(u, v) \neq 0$$

Give explicitly the implicit function  $y = \psi(x)$  that  $f$  defines near the point  $u$  and which satisfies  $\psi(u) = v$ .

### Answer Area

1. Use polar coordinates:  $x = r \cos \theta, y = r \sin \theta$ . Then

$$\frac{\sin(xy)}{x^2 + y^2} = \frac{\sin(r^2 \cos \theta \sin \theta)}{r^2}.$$

For small  $t$ ,  $\sin t \sim t$ , so

$$\frac{\sin(r^2 \cos \theta \sin \theta)}{r^2} \sim \frac{r^2 \cos \theta \sin \theta}{r^2} = \cos \theta \sin \theta.$$

The limit as  $r \rightarrow 0$  depends on  $\theta$ : it equals  $\cos \theta \sin \theta$ , which varies with  $\theta$ . For example:

- Along  $x = 0$  or  $y = 0$ : limit = 0.
- Along  $y = x$ :  $\cos \theta \sin \theta = \frac{1}{2}$  when  $\theta = \pi/4$ .

Since the limit depends on the direction, it does not exist.

2. (a) Compute partial derivatives at  $(0, 0)$ :

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

Similarly  $\frac{\partial f}{\partial y}(0, 0) = 0$ . Thus  $\nabla f(0, 0) = (0, 0)$ .

- (b) For  $(x, y) \neq (0, 0)$ , compute  $\frac{\partial f}{\partial x}$  using quotient rule. Let  $r = \sqrt{x^2 + y^2}$ , then

$$f(x, y) = \frac{x^2 y^2}{r^3}.$$

Compute derivative (or check along a path). Consider approaching  $(0, 0)$  along  $y = x$ : For  $y = x$ ,  $f(x, x) = \frac{x^4}{(2x^2)^{3/2}} = \frac{x^4}{2^{3/2}|x|^3} \sim \frac{|x|}{2^{3/2}}$ . The partial derivative  $\frac{\partial f}{\partial x}$  along this path does not tend to 0 (shows discontinuity). More formally: compute  $\frac{\partial f}{\partial x}(x, y)$  explicitly and show its limit as  $(x, y) \rightarrow (0, 0)$  does not exist or does not equal 0.

3. Since  $f$  is homogeneous of degree  $\alpha$ , Euler's theorem gives:

$$x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = \alpha f(x, y).$$

At  $(u, v)$ ,  $f(u, v) = 0$ , so

$$u \frac{\partial f}{\partial x}(u, v) + v \frac{\partial f}{\partial y}(u, v) = 0.$$

By the Implicit Function Theorem, since  $\frac{\partial f}{\partial y}(u, v) \neq 0$ , there exists  $\psi$  near  $u$  with  $\psi(u) = v$  and  $f(x, \psi(x)) = 0$ .

Differentiate  $f(x, \psi(x)) = 0$ :

$$\frac{\partial f}{\partial x}(x, \psi(x)) + \frac{\partial f}{\partial y}(x, \psi(x))\psi'(x) = 0.$$

At  $x = u$ ,  $\psi(u) = v$ :

$$\psi'(u) = -\frac{\frac{\partial f}{\partial x}(u, v)}{\frac{\partial f}{\partial y}(u, v)}.$$

Using Euler's relation:  $\frac{\partial f}{\partial x}(u, v) = -\frac{v}{u}\frac{\partial f}{\partial y}(u, v)$  (since  $u\frac{\partial f}{\partial x} + v\frac{\partial f}{\partial y} = 0$ ), so

$$\psi'(u) = \frac{v}{u}.$$

The linear approximation near  $u$  is:

$$\psi(x) \approx v + \frac{v}{u}(x - u).$$

For the exact implicit function, if  $f$  is homogeneous, one can show  $\psi(x) = \frac{v}{u}x$  (if  $f$  is linear in  $y$ , but generally it's the tangent approximation).